GOVERNMENT OF WESTERN AUSTRALIA

GSWA Geochronology Record

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Department of Mines, Industry Regulation and Safety

242498.1.pdf *Geochronology Record 1937*

242498.1: quartzite, Mungo Pool

(Youanmi Terrane, Yilgarn Craton)

SS Romano, MTD Wingate, Y LU, IOH Fielding and TJ Ivanic

Location and sampling

MURGOO SG50-14, MURGOO 2143 Zone 50, MGA 433317E 6987137N
Warox Site TJIMUR190009 Sampled on 3 September 2019

This sample was collected from a north-trending outcrop, about 2.4 km south-southwest of Noolya Pool, 2.1 km west of Mungo Pool, and 0.6 km north of the Meeberrie–Wooleen Road.

Geological context

The unit sampled is a quartzite (Fig. 1) near the southeastern margin of the Barloweerie greenstone belt (Baxter, 1974). About 0.1 km to the east-northeast, the quartzite is intruded by a metagranodiorite sheet, which contains 3048–3009 Ma zircon xenocrysts that are essentially identical in appearance, isotopic characteristics, and age to those in the quartzite (GSWA 242497, Romano et al., 2023).

Figure 1. Hand specimen image for sample 242498: quartzite, Mungo Pool

Petrographic description

This sample is a fine-grained, quartz–muscovite rock (Fig. 1) crosscut by quartz veinlets. The quartz–muscovite parts of the sample consist of about 55% quartz, 45% muscovite and trace rutile. Quartz is equant, unstrained, and up to 0.1 mm in diameter. Muscovite laths are subhedral and randomly oriented, and rutile forms trails and small clusters of anhedral crystals up to 0.05 mm long. Quartz grains form a semi-polygonal aggregate with each other and with muscovite laths. The sample contains parallel boudinaged lenses up to 4 mm wide and smaller pods composed of coarsely microcrystalline, unstrained quartz. The lenses and pods are interpreted as remnants of dismembered, tightly folded quartz veins. The quartz–muscovite parts of the sample may represent a metamorphosed quartz-rich siltstone, although the absence of obvious detrital grain shapes in quartz and of sedimentary structures such as bedding, suggest, together with the presence of vein quartz, that the sample could have a hydrothermal origin.

Zircon morphology

Zircons isolated from this sample are pale brown to dark brown and mainly subhedral to euhedral. The crystals are up to

200 mm long, and equant to slightly elongate, with aspect ratios up to 3:1. In cathodoluminescence (CL) images, broad concentric zoning is common, although poorly defined. A CL image of representative zircons is shown in Figure 2.

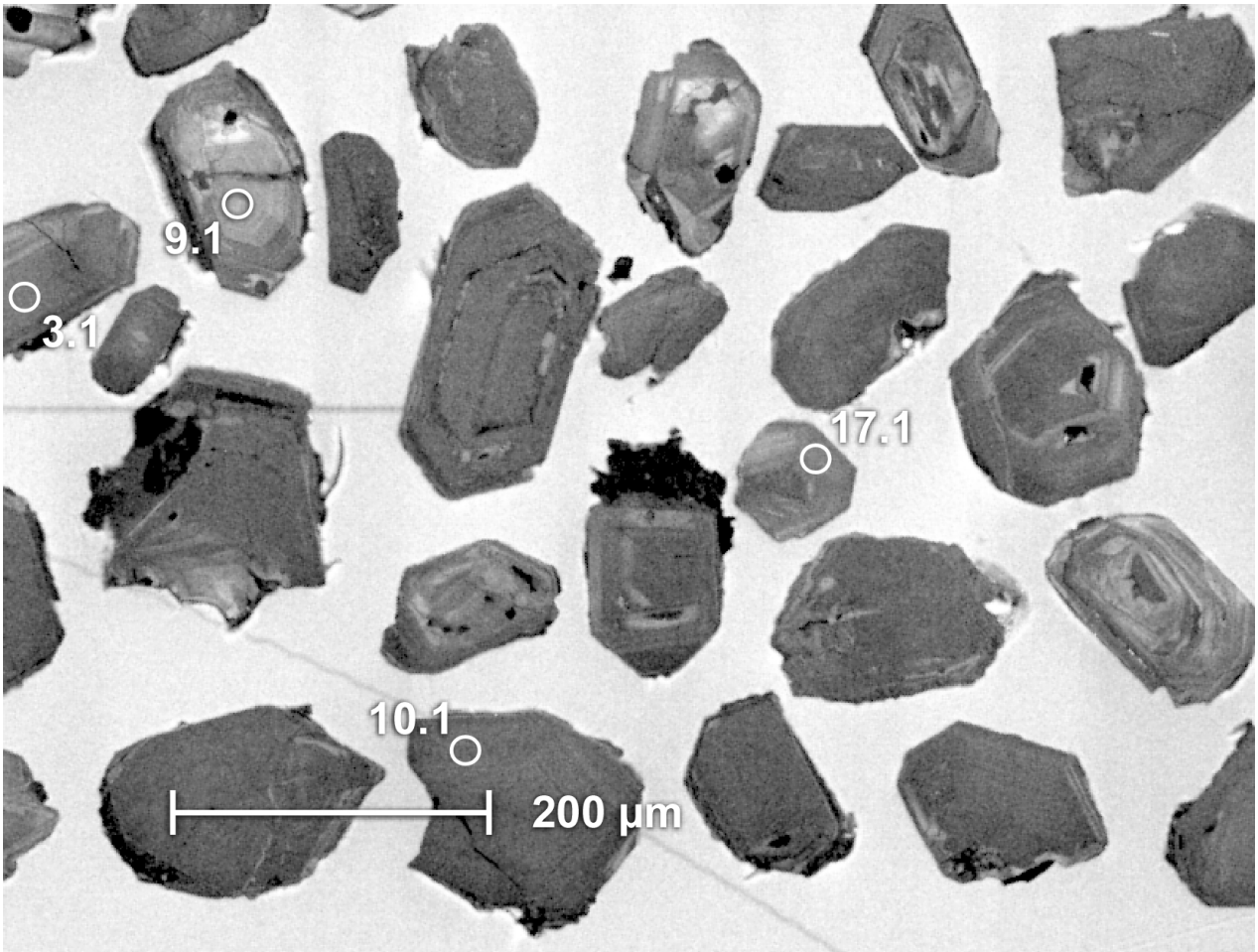

Figure 2. Cathodoluminescence image of representative zircons from sample 242498: quartzite, Mungo Pool. Numbered circles indicate the approximate locations of analysis sites

Analytical details

This sample was analysed on 16–17 September 2021 using SHRIMP-B. Ten analyses of the M257 standard were obtained during the session, of which nine analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.18% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1 (provided in the Links section). Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Forty-two analyses were obtained from 41 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). Five analyses are >5% discordant. The dates obtained from these five analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 37 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group Y comprises one analysis (Table 1), which yields a $^{207}Pb/^{206}Pb^*$ date of 3019 ± 8 Ma (1σ).

Group S comprises 36 analyses of 36 zircons (Table 1), which yield $^{207}Pb/^{206}Pb^*$ dates of 3056–3019 Ma.

It is possible that all of the analyses in Group Y and S are of unmodified detrital zircons, in which case the date of 3019 ± 8 Ma (1σ) for the single analysis in Group Y represents a maximum age of deposition for the sedimentary precursor. A more conservative estimate of the maximum depositional age can be based on the weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 3037 ± 2 Ma (MSWD = 1.2) for all 37 analyses in Groups Y and S.

The data for combined Groups Y and S indicate a single significant age component at c. 3037 Ma, based on contributions from 37 analyses. This is interpreted as the age of the dominant zircon-crystallizing rock in the detrital source region, or the age of a detrital component within sediments that have been reworked into this rock.

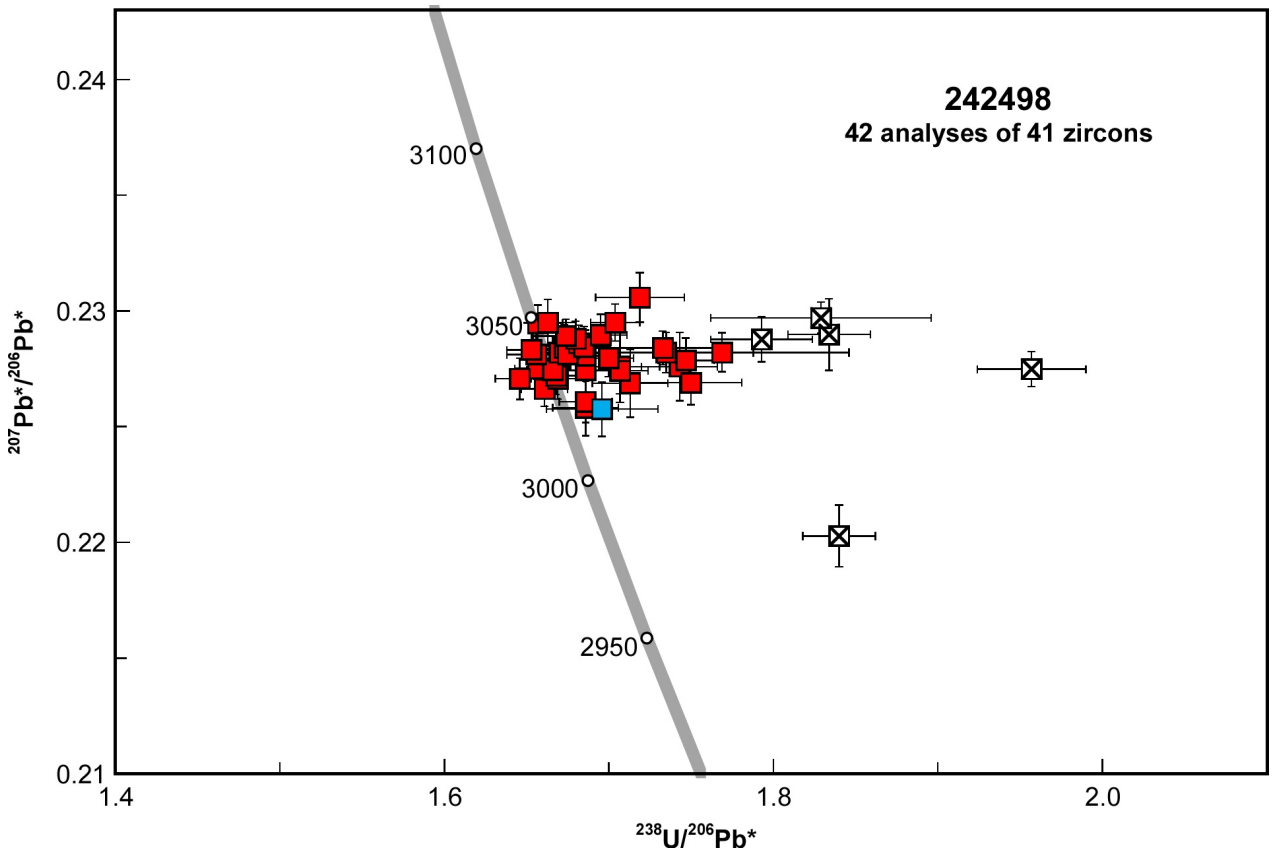

Figure 3. U–Pb analytical data for zircons from sample 242498: quartzite, Mungo Pool. Blue square indicates Group Y (youngest detrital zircon); red squares indicate Group S (older detrital zircons); crossed squares indicate Group D (discordance >5%)

Links

Table 1: Ion microprobe analytical results for zircons from sample 242498: quartzite, Mungo Pool: 242498.1.Table1.xls
SHRIMP spot images: 242498_spots.pdf
Introduction to geochronology information : Intro_2023.pdf

References

Baxter, JL (compilers) 1974, Murgoo, Western Australia: Geological Survey of Western Australia, 1:250 000 Geological Series Explanatory Notes, 21p. View Reference online

Romano, SS, Lu, Y, Fielding, IOH, Wingate, MTD and Ivanic, TJ 2023, 242497: metagranodiorite, Mungo Pool; Geological Survey of Western Australia; Geochronology Record 1936: Geological Survey of Western Australia, <www.dmirs.wa.gov.au/geochron>.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Recommended reference for this publication

Romano, SS, Wingate, MTD, LU, Y, Fielding, IOH and Ivanic, TJ 2023, 242498.1: quartzite, Mungo Pool; Geochronology Record 1937: Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract viewed 13 December 2023. <www.dmirs.wa.gov.au/geochron>

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nearest 100 m.

Capitalized names in text refer to standard 1:100 000 map sheets, unless otherwise indicated.

WAROX is GSWA's field observation and sample database. WAROX site IDs have the format 'ABCXXXnnnnnnSS', where ABC = geologist username, XXX = project or map code, nnnnnn = 6-digit site number, and SS = optional alphabetic suffix (maximum 2 characters).

Isotope and element analyses are routinely conducted using the GeoHistory laser ablation ICP-MS and Sensitive High-Resolution Ion Microprobe (SHRIMP) facilities at the John de Laeter Centre (JdLC), Curtin University, with the financial support of the Australian Research Council and AuScope National Collaborative Research Infrastructure Strategy (NCRIS), and the Microscopy Australia laser ablation ICP-MS facility at the Centre for Microscopy, Characterisation & Analysis (CMCA), The University of Western Australia (UWA). The TESCAN Integrated Mineral Analyser (TIMA) instrument was funded by a grant from the Australian Research Council (LE140100150) and is operated by the JdLC with the support of the Geological Survey of Western Australia, UWA and Murdoch University. Mineral analyses are routinely obtained using the electron probe microanalyser (EPMA) facilities at the CMCA at UWA, at Adelaide Microscopy at the University of Adelaide and the Central Science Laboratory at the University of Tasmania.

Digital data related to WA Geology Online, including geochronology and digital geology, are available online at the Department's Data and Software Centre and may be viewed in map context at GeoVIEW.WA. For further information about this Geochronology Record, please contact geochronology@dmirs.wa.gov.au and quote the geochronology record and/or the sample number.

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